

The Chinese Housing Market and Labor Supply – A Literature Review

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I. Introduction

It has been empirically shown that there is a causal relationship between the housing market and labor supply both on the extensive i.e. someone's decision to work or labor force participation (LFP) and intensive margin i.e. someone who does participate in the labor forces' decision of how many hours to work or hours decision (HD). At its core, the question can be distilled to this: if household wealth causally influences an individual's decision to work or an individual's hours decision, in which way is this the case?

Over the past two decades, China has become an ideal environment to study this causal relationship between housing and labor supply. Due to soaring home prices and the implementation of a 1994 housing reform China offers a rich environment for generating meaningful empirical results.

In this literature review, I examine the emerging literature on the housing market and its relationship with labor supply in China. First, I will discuss background establishing the relationship between the housing market and labor supply, discussing related papers and established theory. I will discuss the factors that make China a proper candidate for this research i.e. the 1994 housing reform. Followed by the empirical results of the literature, research methods and how the literature handles the issues of endogeneity. Finally, the literature review ends with a discussion involving the differences between men and women and other topics.

II. Background and Theoretical Framework

Several mechanisms have been proposed in the literature to explain how housing markets, particularly housing price changes, influence labor supply deci-

sions. These mechanisms operate through wealth effects, budget constraints, regional mobility, and behavioral responses. In this paper, most of the research is focused on wealth effects. Wealth effects are the effects observed when increased wealth due to increased value of a home may change someone's decision to work. Homeowners may feel less pressure to work as their wealth increases, particularly if they anticipate being able to sell their home at a profit or borrow against its increased value. The theory that we are working with is relatively simple. When home prices go up, people who already own homes feel wealthier. When people feel wealthier, they might feel like they do not need to work as much or at all. This is known as wealth effects.

Housing booms provide a natural experiment framework to examine the potential causal relationship between the housing market and labor supply. These economic shocks introduce measurable variations, enabling researchers to analyze changes. An existing literature suggests people change labor supply decisions given the status of household wealth. Take the housing boom in the US prior to the 2008 housing crisis, Charles et al. (2018) show that people on the margin of making the decision to go to school or go work changed their LFP decision (labor supply decision on the extensive margin) – the result in the paper shows a drop in college attendance for 2 year schools prior to the 2008 housing crisis, followed by a return to trend after the 2008 financial crisis - on the surface this could be seen as a way the housing market influences someone's LFP decision. Furthermore, Cesarini et al. (2017) shows that there is a direct effect of wealth on labor supply. They show that winning the Swedish lottery significantly reduces earnings – with this reduction being immediate and persistent. This result shows a clear relationship between wealth shocks and LFP.

III. Chinese Housing Boom and 1994 Housing Reform

Chen et al. (2017) characterizes “The Great Housing Boom of China” as a series of paradoxical events. The housing market in China has been growing faster than national income over the past decade. With income growth at 11 percent and a return on investment of housing at 17 percent, these conditions suggest a potential for unsustainable growth. Between the years 1998 and 2012, the return to capital (factoring in depreciation and inflation – in real terms) was 20 percent or above over a 14- year period. Home vacancy rates were approximately 22.4 percent in 2013. Despite this, investment in housing continues to grow – due to optimistic outlooks on future demand for housing. In general terms, if housing prices continue to grow over income – housing in China could soon be considered a form of debt rather than an asset. This could in theory drive labor supply through debt driven labor supply.

A housing reform took place in China which allowed for previously state-owned housing to be purchased by previous tenants at heavily subsidized prices. This reform per Chen et al. (2023) made China a country with one of the highest home ownership rates in the world, and as we will show later this reform significantly reduced labor force participation (LFP). Prior to the housing reform according to Li et al. (2014) a couple could apply to acquire housing through their employer, and after the reform they could purchase commercial houses from the market. The ability to acquire housing in a different way than was historically available shifted the group of people who would be able to own housing. This fact is exploited in the research that follows.

In our discussion it may be important to note that China and the U.S. differ in meaningful ways. The way in which “housing” is meant is completely different, in the U.S. there are homes with a yard that are all separated. Or in some cases people who own condos may consider this a home. While in China – and the literature is not completely clear on this – may define homes more so like apartments. Where our observation of “homes” in China may be strictly different of what we think of in the United States. China also has various other differences when it comes to measuring how the housing market impacts labor supply. For example, China is a bigger country that has a planned economy and is more densely populated. Also, the expectations of ones’ job that people have in China may differ also - for example people in China tend to move to areas specifically for the location of where they intend to work like people in

the U.S. however, since loyalty is seen as very important these moves to certain areas tend to last a longer period of time rather than being able to move from place to place depending on the job - this emphasizes the importance of owning a home in an area where someone has a job. These differences may affect the results we see in the empirical literature that we observe. Furthermore, the housing market in China and the housing market in the United States may not have the same exact causal relationship between housing and labor supply for these reasons(i.e. the mechanisms that relate the two). Housing may have different effects on Chinese and U.S. labor supply respectively.

IV. Empirical Results from the Chinese Housing Market

This literature review examines the relationship between the housing market and labor supply in China, with a focus on the housing boom and the 1994 housing reform. These events provide a basis for measuring how changes in the housing market affect labor supply decisions, which are typically categorized into two margins: the intensive margin (how many hours to work or the hours decision) and the extensive margin (whether to participate in the labor market at all also known as labor force participation). The review begins with research focused on labor supply on the extensive margin (labor force participation) followed by research focused on labor supply on the intensive margin (hours decision).

Labor Supply: Extensive Margin

Much of the literature demonstrates a negative relationship between housing prices and LFP, indicating that rising housing wealth reduces LFP via wealth effects. For instance, leveraging the 1994 housing reform, Chen et al. (2023) finds that those affected by the reform were 15.1% less likely to participate in the labor force. Notably, women were 18.9% more likely to drop out of the labor market entirely, while men were 10% less likely to participate. The 18.9% drop in female participation was calculated from two marginal effects (see Chen et al (2023), p. 230).

Fu et al. (2016) observe that a 100,000-yuan increase in household value reduces the likelihood of female homeowners participating in the labor force by 1.37% and increases the probability of becoming a housewife by 1.49%. Conversely, they find little effect on men’s LFP.

Zhao et. al (2017) finds itself in contrast to Fu et al. (2016) and Chen et al. (2023). Zhao et al. (2017) investigates the effects of house price appreciation on homeowners' LFP. They find that a 100,000 yuan increase in housing wealth increases the probability of participating in the labor force by 1.9%. A rationale is also developed in the paper that suggests as home prices appreciate, people in China (who value housing as an asset for an expanding family and securing homes for their family members) will pursue work in a housing market with homes that appreciate – thus securing housing for loved ones.

Labor Supply: Intensive Margin

Now transitioning to the research regarding labor supply on the intensive margin. Jiang et al. (2022) use self-reported housing wealth and relative housing wealth to measure the effects on labor supply on the intensive margin (hours decision). They find that relative housing wealth reduces homeowners' working hours, though regional housing wealth has no statistically significant impact on working hours. This discrepancy suggests that attitudes about personal wealth status influence the decision of how many hours to work, rather than simply the true increase in value of one's home. Using an IV approach, they estimate that a one-million-yuan increase in relative housing wealth reduces annual working hours by 3.81% equivalent to about nine fewer working days per year. With a further reduction in female working hours by 7.06% reduction given a one-million-yuan increase. Several robustness checks are used to verify these results.

Building on this, Jiang et al. (2023) refine the measurement of relative housing wealth by comparing self-reported housing values to reference values derived from a hedonic model. They again find a negative effect on the intensive margin, particularly for women. Specifically, a one-million-yuan increase in relative housing wealth leads to a 4.34% reduction in annual working hours (approximately 73 hours) among women. When using their refined IV measure, this reduction is slightly lower at 8.65% (approximately 18 working days). These findings suggest that relative housing wealth plays a dominant role in explaining the decline in female homeowners' working hours.

Additionally, Jiang et al. (2023) contribute a novel insight: *perceived differences* in housing wealth influence labor supply decisions. While this result may invite broader sociological discussion, this review focuses on the empirical findings, highlighting how shifts in housing wealth impact labor supply de-

cisions, particularly among women.

V. Research Methods and Identification Strategy

Instrumental Variables (IV)

Measuring the causal relationship between the housing market and labor supply is complicated by endogeneity. For instance, individuals aiming to purchase expensive homes may already have high-paying jobs, while those with limited skills or work hours may be less likely to buy such homes. This makes it difficult to disentangle whether housing wealth affects labor supply or vice versa.

To address this, researchers use instrumental variables (IV) to isolate exogenous changes in housing wealth. For example, house price appreciation or purchase prices can serve as instruments to capture how changes in home values influence decisions to work. The goal is to measure the true causal effect of housing wealth on labor supply rather than confounding factors, such as pre-existing employment conditions or wealth aspirations.

While IV approaches help identify causal effects, they are not without limitations. Variations in instruments may subtly shift the target being measured, complicating the interpretation of results. Despite these challenges, existing studies support the theoretical link between housing wealth and labor supply while highlighting heterogeneity across income levels, regions, and demographics. The remaining papers referenced in the literature review implement an instrumental variable methodology to account for endogenous regressors. Zhao (2018) conduct an IV probit model account for their endogenous regressor of house price appreciation. This is endogenous in the sense that house price appreciation may be related to the types of areas that already have people who work more or at higher paying jobs. This may suffer from reverse causality as labor supply and house price appreciation may influence one another. The instrumental variable deployed in this case is the price at which the home was purchased at. Home purchase price is correlated with the household appreciation but is uncorrelated with the current labor supply decision.

There are two specifications from Jiang (2022) followed by a third mentioned in Jiang (2023) that each deploy a subtly different IV. Specification 1 of Jiang (2022) deals with the endogenous regressor of self-reported housing wealth by deploying the IV defined as average self-reported housing wealth for all house-

holds excluding the homeowner’s house. The IV is correlated with the homeowners housing wealth and is not correlated with the unobserved attributes of homeowners that potentially influence labor supply, thus is argued to be valid. The second specification uses relative housing wealth as the endogenous regressor and similarly uses the IV as before, deploying the average relative housing wealth for all households excluding the homeowner. Jiang (2023) attempts to measure something very similar, working with the endogenous regressor as relative housing wealth again but this time deploying the IV defined as the average relative housing wealth of other homeowners living in the same community in each year.

Fu (2016) deals with the endogenous regressor of total housing wealth change and uses the IV defined as average total housing wealth change in other households of the same community. This is correlated with the housing wealth change of an individual but not correlated to the LFP decision of that person.

While many of these IV’s are very clever and rather nuanced (at least for this literature), it does not seem clear in all these cases that these IVs that are deployed in the literature isolate the wealth effects of households on labor supply strictly.

Difference-in-Differences (DID)

Difference-in-differences (DID), is a statistical method used to estimate causal effects by comparing the changes in outcomes over time between a treatment group and a control group. It assumes that, in the absence of treatment, the two groups would have followed parallel paths in their outcome variables. Chen (2023) uses a DID method using CHNS data, with the treatment group being defined as individuals who were impacted by the 1994 housing reform (that is people who resided in public housing that were then given the opportunity to purchase these homes). The control group were households that were already privately owned and not impacted by the housing reform. In general this strategy would be more promising than most of the IVs previously mentioned. However since there are not many ”surprise” treatments like the 1994 housing reform, it is not at all surprising that this is one of the only papers utilizing this method.

A summary of empirical results with IVs used and descriptions of the treatment and control group for the DID method can be found on the next two pages.

Table of Results and Instrumental Variables

Paper	Primary Empirical Result	Data	Methodology	Endogenous Regressor	Instrumental Variable
Zhao (2018)	Extensive Margin: 100,000 yuan increase in housing value increased probability of LFP by 1.9%	CFPS	IV Probit	House Price Appreciation	House Purchased Price
Jiang (2022) Specification 1	Intensive Margin: 1 million yuan increase in housing value reduces working hours by 3.81% (9 working days) and reduces women's working hours by 7.06%	CHNS	2SLS	Self-Reported Housing Wealth	Average self-reported housing wealth for all households excluding the homeowner's house
Jiang (2022) Specification 2	—	CHNS	2SLS	Relative Housing Wealth	The average relative housing wealth for all households excluding the homeowner
Jiang (2023)	Intensive Margin: A one million yuan increase in housing value reduces working hours by 8.65% for women (18 working days) using IV estimate	CHNS	2SLS	Relative Housing Wealth	The average relative housing wealth of other homeowners living in the same community in each year
Fu (2016)	Extensive Margin: 100,000 yuan increase in household value reduces likelihood of LFP for women by 1.37% and increases the likelihood of becoming a housewife by 1.49%	CHFS	2SLS and IV Probit	Total Housing Wealth Change	Average total housing wealth change in other households of the same community

Table of Results from Difference in Differences Method

Paper	Primary Empirical Result	Data	Methodology	Treatment Group	Control Group
Chen (2023)	Extensive Margin: 15.1% drop in LFP and 18.9% drop of LFP for women	CHNS	Difference in Differences (DID) estimated by probit regression	Individuals who reside in public housing (rented from state employers) and have at least one family member working in government, state institutes or state-owned enterprises before the housing reform.	The comparison group that includes households that reside in rented private-owned house, or households that live in public houses but work in private sectors before the housing reform.

VI. Differences Among Men and Women

The literature highlights significant differences between men and women in their labor supply decisions in response to changes in the housing market. A chapter from Contemporary Labor Economics (Ninth Edition), titled The Theory of Individual Labor Supply, notes that "male labor supply is quite insensitive to changes in wage rates, whereas female labor supply is fairly responsive to changes in wage rates." This observation underscores the gendered dynamics of labor supply, which are further reflected in housing market trends and their influence on labor supply decisions among men and women.

Research by Wu et al. (2015) illustrates how Women's Labor Force Participation (WLFP) in China underwent dramatic changes during the 1990–2010 period. The authors find that WLFP decreased significantly from 1990 to 2003 due to institutional transformations, such as economic restructuring and the decline of guaranteed employment in state-owned enterprises. Post-2003, however, WLFP increased, largely driven by rising living costs, potentially influenced by the soaring housing prices during this period. This suggests that women's labor supply decisions are particularly sensitive to external economic pressures, including housing affordability and family income dynamics.

These findings are consistent with the broader economic theory of labor supply, where women, often seen as secondary earners, tend to adjust their labor supply more readily in response to changes in household income or living expenses. In the context of the housing market, women's labor supply appears to be more elastic, as they may enter or exit the labor force depending on the family's financial needs during periods of housing booms or busts.

Bertrand et al. (2021) provide another perspective on the intersection of gender, labor, and social dynamics by examining the marriage gap among skilled and unskilled women in developed countries. The study finds that skilled women marry at lower rates, a trend shaped by improved labor market opportunities for women and negative social attitudes toward working women. These findings have implications for the relationship between housing markets and WLFP: as skilled women increasingly prioritize career opportunities, their responsiveness to housing market shocks may differ from that of less skilled women, whose decisions are more constrained by traditional family roles or social norms. This is further backed up by the explicitly stated empirical result from Fu et al. (2016) that observe an increase in the

likelihood of becoming a housewife by 1.49% when the housing market goes up.

The gender differences in labor market responsiveness have important implications for both the extensive and intensive margins of labor supply. Men, who are often viewed as the primary breadwinners, are less likely to withdraw entirely from the labor force (extensive margin) in response to changes in housing wealth or market conditions. Instead, they may adjust their working hours or effort (intensive margin). Women, however, are more likely to adjust their participation on the extensive margin, such as by entering or exiting the labor force altogether. This is consistent with traditional family dynamics, where women may prioritize caregiving or household responsibilities, especially in dual-income households where rising housing wealth reduces the need for a second income.

Rising housing prices create additional pressures that affect men and women differently. For men, increased housing costs may lead to longer working hours or taking on multiple jobs to meet financial obligations. For women, the decision to work may hinge more on household income dynamics. If housing wealth increases significantly, women in dual-income households may opt out of the labor force, while those in lower-income households may be compelled to join the labor force to meet financial demands. This dual effect highlights the complex ways in which gender roles mediate the relationship between the housing market and labor supply.

Overall, the gendered differences in labor supply responsiveness to housing market shocks reflect a confluence of economic, social, and cultural factors. Women's labor supply tends to be more elastic, influenced by family income, social norms, and caregiving responsibilities. Men's labor supply, on the other hand, is more inelastic, driven by their traditional role as primary earners. These differences underscore the importance of considering gender when analyzing the causal relationship between housing markets and labor supply, as well as the broader implications of housing policies on workforce dynamics.

VII. Final Considerations

In summary, the above literature review specifically looks at literature discussing the impact in China of the relationship between the housing market and labor supply. Perhaps some obvious discussion is required to draw out some meaningful points from what we have seen so far. The way in which the effect of the housing market on labor supply is measured in

some ways makes it unclear if there is a direct mechanism (a true causal function/mechanism) that relates these two things (perhaps it is a random observation and the two things are not truly related). Perhaps the mechanism/mechanisms we are looking for are indirect means which could be more clearly measured in the future with more precise data and state of the art methods. In any case – methods like instrumental variables and difference in differences provide at least a solid theoretical framework to establish such a relationship for the time being.

Other theoretical mechanisms to consider in a broad scope among this topic could be how unemployment also plays a role within the changing housing market. Oswald et al (1996) argues that high unemployment can be explained by a decline in the private house-rental market and an increase in homeownership. With a result of a 10% increase in homeownership causing a 2% increase in unemployment. This result is driven by homeowners who are not able to move from place to place as easily as renters for different job opportunities, thus creating a wave of unemployment. Further on the topic of unemployment and how it also relates to the housing market, Farber et al (2012) finds that after a housing crisis in 2008 – unemployment spells tend to last longer than ever (including after the period of the great depression). In this case the cause being economic immobility that makes it more difficult to sell one’s house – thus making it difficult to take new jobs in the same way Oswald et al (1996) found. Finally, Blanchflower et al (2013) finds that rises in the homeownership rate in a U.S. state are a precursor for an eventual sharp increase in unemployment in that state. If we were to apply these result to China – with such an increase in homeownership we could forecast a potential rise in unemployment for various reasons stated previously, economic immobility, inability to sell ones home and longer periods than previously observed spells of unemployment. It also puts into question how unemployment is being measured in the literature we have discussed (if at all) and how these sorts of questions could be explored in further research.

Policy makers may find this type of research useful in a variety of ways. From the standpoint of employment and expected number of hours worked per employee – we could implement various policies that would react to shocks in the housing market that have an impact on labor supply. If say, a policy maker wished to maximize employment both extensively and intensively – there would be incentives to stabilize the growth of the housing market to manageable levels. The opposite would also be true that if there was a particular negative shock to the housing market, we could expect people to seek more job opportunities and depending on the job market at the time this could also increase unemployment. As a result, a combination of falling housing prices would be in theory coupled with increased unemployment. Given this result, the government may want to intervene in some ways to stabilize house value and employment rates.

Chen et al. (2023) highlights the importance of aligning housing reforms with policy goals related to labor force participation (LFP). Since different countries have varying optimal LFP targets, housing policies can serve as instruments to stabilize economies by influencing LFP and unemployment. For example, making housing more accessible can increase wealth for previously excluded groups, potentially reducing their labor supply. Conversely, restricting access or lowering home values could incentivize higher labor force participation if the relationship between housing and LFP is positive.

Lastly, in terms of a discussion of a collapse of China’s housing bubble, with rapidly falling home values, could significantly impact labor supply. Many individuals may choose to work more hours or re-enter the workforce, potentially affecting home-based activities like childcare, household management, and human capital accumulation. This shift in labor supply could lead to job shortages, increased competition, and unemployment. In many ways, China could face a scenario similar to the 2008 U.S. housing crisis, resulting in a sharp short-term decline in economic welfare.

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